

Robotics Maze G1 – Interview Project Summary

Project Goal

Build a reproducible robotics simulation around a Unitree G1 humanoid navigating a maze. The goal was to demonstrate robotics engineering practices including simulation, planning, testing, ROS2 integration, SLAM, and autonomous navigation.

Development Strategy

The project was built milestone-by-milestone. Each milestone isolated a specific risk so debugging remained manageable and subsystem behavior could be validated before adding complexity.

Maze Generation

A seeded randomized DFS generator creates deterministic mazes. BFS validation proves solvability and produces a shortest path through the maze. Deterministic seeds make failures reproducible.

2D to 3D Conversion

The maze is first represented as a logical 2D grid. Wall cells are converted into MuJoCo geometry, creating a physical world while keeping maze logic independent of simulation.

Oracle Navigation

An oracle baseline uses MuJoCo ground-truth pose. This validates planning and control without localization uncertainty and serves as a debugging baseline.

Path Planning and Following

The planner computes routes and waypoints. The follower computes heading and distance errors, applies velocity limits and tolerances, and generates robot commands. Arc turns and recovery logic were added to improve behavior.

Docker and ROS2

A portable Docker environment with ROS2 Humble, Nav2, SLAM Toolbox, and visualization tooling was created to support reproducible execution across platforms.

Testing Philosophy

Smoke tests verify startup and configuration. Unit tests validate maze generation, planning, world generation, controller behavior, ROS integration assumptions, Docker support, and visualization artifacts.

Nav2 and SLAM Roadmap

Phase 5 validates SLAM, costmaps, planners, controllers, BT Navigator, and cmd_vel generation. Phase 6 connects Nav2 output to the Unitree locomotion policy and closes the navigation loop.

Key Challenges

Understanding BFS vs DFS, converting a logical maze into a physical world, designing an oracle navigation baseline, integrating ROS2/Nav2, handling humanoid locomotion constraints, and creating a maintainable testing strategy.

Interview Defense

Emphasize architecture, subsystem isolation, reproducibility, testing, debugging methodology, and staged integration. The strongest point is the engineering process rather than a single demo result.